GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. For known prepared positive causal instruments with a finite separating affine family of executable tests, we prove a matching finite-state risk law in a genuinely executed geometric-stop experiment. Preparation-injected calibration identifies the exact stopped physical–observer ensemble by a positive resolvent, without stationary initialization or an observer mixing assumption. Its regret against all full-history adaptive acquisition strategies is exactly an acquired conditional Jensen gap of a discounted Bellman potential. Actual acquisition mass, global cells and feedback-uniform raw overlap make this gap comparable to $M^{-2/d}$ uniformly over every finite mean acquisition budget. An initial label and internal halting labels are explicitly charged. A hard cap has a separately proved tail term. Controlled Gaussian sensors and noncommuting quantum instruments verify the same theorem from raw kernels; marked acquired-rank changes retain their actual masses. An explicit binary code, attained calibration–deficiency alternatives, and finite-precision and simulator accounting complete the same-task transfer. The theorem optimizes acquisition, not the imposed stopping rule, and does not assume a pre-existing statistical manifold.

1. The mother problem and the first executed run

A causal experiment supplies histories, not a manifold. Its geometry begins with the predictions of future experiments that can actually be performed. Even when finitely many such predictions form a sufficient state, a finite observer cannot retain an arbitrary point of that state space. Its actions determine the reports it will receive; its compression changes subsequent actions; and histories assigned the same label must be represented by one executable continuation. A risk bound must therefore concern the distribution acquired by that program, not only the set of formally reachable predictions.

A second distinction is equally important. An invariant calibrated observer is not automatically calibrated when started from the stated preparation. Converting an invariant assertion into a first-run assertion requires either an entrance estimate or a direct identification of the executed occupation law. The present article follows the second route. It proves a resource law for an independent geometric stopping experiment, whose mean number of acquisitions is finite. The stopping experiment is performed, rather than used as a formal abbreviation for an infinite-horizon cost.

The central construction has a dedicated initial label. The remaining labels are common cells of the predictive test image. Their conditional means are calibrated to the distribution at the actual stopping call. The calibration equation includes the preparation as an inhomogeneous term. Once the table has been fixed, its unique solution is the positive resolvent of the executed physical–observer instrument. Thus a fixed point in the construction is not an assertion that the machine starts in equilibrium. A discounted Bellman identity then cancels the continuation contribution and identifies the entire regret with one conditional Jensen gap under the actually acquired row mixture.

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Under the experiment-level certificates in Definition 3.1, Theorem 3.2 gives

$$\mathcal{R}_{\mathrm{cp}}(s, M; u_0) - V_s(u_0) \asymp \mathcal{R}_{\mathrm{hom}}(s, M; u_0) - V_s(u_0) \asymp M^{-2/d},$$

uniformly for $0 < s \le 1$ and $M \ge 2$. Here s is the prescribed independent stop probability, $1/s$ is the mean number of actual acquisition calls, and V_s is the optimum over all full-history adaptive acquisition strategies in the same task. A completely autonomous implementation, including its halt status, uses at most twice the recurrent labels and satisfies the same order. The parameter d is verified in the executable test image by actual acquired densities; it is not the dimension of an arbitrary parametrization of the experiment.

The exact calibration identity requires no feedback overlap. That certificate is used only to keep the Bellman oscillation, and hence the rate constant, uniform as the mean number of calls increases. No convergence or spectral gap of the finite observer is used. The table may depend on the preparation and on s: the assertion is uniform bounds, not one common table for all preparations and hazards. The finite-acquisition, uniform-audit and stationary results retained in the companion have different protocols and remain separate theorems [13].

The raw realizations are controlled Gaussian observations and noncommuting quantum instruments. Both verify the same theorem before a code or budget is selected. A visible conditional expectation in the quantum instrument yields a changing acquired-rank extension. The lower retains the actual weight of every component. Geometric stopping is not arbitrary optimal stopping, and a finite mean budget is not a hard cap; Theorem 5.1 proves the capped comparison explicitly.

Conditional projection, discounted dynamic programming, positive resolvents and Kakutani's theorem are standard tools. The new assertion proved here is their preparation-injected causal calibration, the exact stopped regret identity, and its hard-state realization with matching acquired-mass lower bounds. Value-directed compression already relates belief approximation to decisions [4]; finite-memory control approximation and learning have a substantial literature [7, 10, 5]. Section 10 distinguishes their memory models and guarantees from this theorem. Neither the name of the program nor the preservation of earlier papers is a mathematical novelty claim.

2. Experiments, predictive realization and stopping

2.1. **Raw objects and executable tests.** All history, preparation, hidden-state and report spaces are standard Borel. The finite nonempty action set A is the same at equivalent histories; any action restriction or public resource record belongs to the interface being compared. Raw prepared kernels are positive and normalized. Actions use only already available history and independent private randomization. No program reads an unknown physical parameter. The quantitative theorem below concerns a known instrument and known preparation; the morphism theorem separately quantifies over parameter families.

Fix legal finite future experiments and bounded terminal readouts, closed under their legal one-step continuations. A finite separating affine test realization means the following. The constant test and a finite vector $u(h)$ of executable-test expectations are Borel, their attained image has a Borel realization section, and equality of these coordinates is equivalent to equality of all stipulated future tests. Realizable random mixtures form a compact convex base $B \subset \mathbb{R}^d$, full-dimensional in this test chart. For every action the joint unnormalized expectation of each coordinate test and the next report is affine in the incoming coordinates. These are hypotheses to be verified, not consequences of positivity alone.

Write $C = \{w(1, u) : w \ge 0, u \in B\}$ and $e(w, v) = w$. The affine closure is represented by integrable positive linear instruments $L_a(y)$ on this finite cone, relative to reference probabilities λ_a; $\int L_a \, d\lambda_a$ preserves e. Define

$$L_a(y)(1, u) = (\ell_a(y, u), U_a(y, u)), \qquad F_a(y, u) = U_a(y, u)/\ell_a(y, u). \tag{1}$$

All raw identities hold off one reference-null set for each action, simultaneously for every u. If the likelihood is zero, positivity and bounded coordinates force $U_a(y,u)=0$; that event contributes zero without division. The actual acquired row and its operator are

$$\mu_{a,u}=(F_a(\cdot,u))_{\#}(\ell_a(\cdot,u)\lambda_a),\qquad K_af(u)=\int f(v)\mu_{a,u}(dv). \tag{2}$$

The realized test state is not supplied as a costless register to a finite observer.

Lemma 2.1 (Measurable predictive realization)**.** *The attained test image realizes the predictive quotient. It is minimal among sufficient Borel statistics whose attained images have Borel realization sections. The report and update kernels descend jointly in the finite action variable.*

Proof. The separating hypothesis identifies precisely the equivalence classes. Compose a realization section of another sufficient statistic with $h\mapsto u(h)$; sufficiency makes the value independent of the section representative. This gives Borel factorization. Formula (1), with the common null sets, gives the descended kernels. Successive integration realizes all legal finite continuations. This proves minimality in the stated category; it does not claim that an arbitrary measurable quotient is standard Borel, or that B is an acquired support. □

2.2. **One protected audit.** A fixed executable audit Y^{aud} has conditional mean u and affine second moment $c(u)$. A response $r\in B$, committed before the audit, has loss

$$\mathbb{E}[\|Y^{\mathrm{aud}}-r\|^2\mid u]=c(u)-2\langle u,r\rangle+\|r\|^2,\qquad g(u)=c(u)-\|u\|^2. \tag{3}$$

Assume $0\le\|Y^{\mathrm{aud}}-r\|^2\le L_0$. The actual audit noise remains in c. For a categorical audit one uses its affine probability hyperplane with the induced Euclidean metric. For quantum experiments only this final audit is executed; no simultaneous intermediate measurement of incompatible tests is assumed.

Fix $s\in(0,1]$ and put $\beta=1-s$. After each acquisition, and after the finite update but before the audit, an independent fresh coin terminates with probability s. Thus

$$\mathbb{P}(\tau=t)=s\beta^{t-1},\quad t\ge1,\qquad \mathbb{E}\tau=s^{-1}. \tag{4}$$

The hazard is independent of the physical state, reports and observer randomizations, conditional on every pre-coin history. Until the coin terminates, it conveys only that the run continues. The process is stopped and audited once. An unaudited reference execution is only a coupling device for these finite stopped laws. There is no state-dependent selection of favorable stopping reports.

For a full-history strategy π, let X_t be its exact predictive state and set

$$\nu_s^\pi=s\sum_{t\ge1}\beta^{t-1}\,\mathrm{Law}_\pi(X_t),\quad B_s^\pi=\int g\,d\nu_s^\pi,\quad V_s(u_0)=\inf_\pi B_s^\pi. \tag{5}$$

These are probability laws and risks of the actual stopped experiment. The controller may adaptively choose each acquisition action, but cannot choose the imposed hazard.

A checkpoint receives the full history at the request and maps it to at most M labels, with one label-only decoder. Its earlier acquisition workspace is not counted as M. A homogeneous observer has at most M total labels, an initial label, fixed action probabilities p_s, one Borel report-update row at each label, and a fixed readout for each label. Only that label is retained between calls. An external stop request reads it. Its infimum risk is $\mathcal{R}_{\mathrm{hom}}(s,M;u_0)$.

For the internally stopped architecture, $\mathcal{R}_{\mathrm{int}}(s,M;u_0)$, halt status is part of the label. At each report the same prescribed independent coin selects halt versus continuation; no external observer clock is supplied. Halt labels have no further acquisition action. Section 5 charges their separate copies. The complete call count in either core architecture is $\tau+2$ including initial preparation and final audit; hence its mean is $s^{-1}+2$ and it has no finite deterministic cap. Program constants, report precision, workspace and physical time are distinct from label count.

3. A preparation-injected acquired-geometry theorem

Definition 3.1 (Geometrically regular controlled instruments)**.** In Section 2, let $K \subset \operatorname{int} D \subset B$ be compact convex sets with $K + B_r \subset D$ for some $r > 0$. The preparation u_0 lies in D. Require the following certificates before a program, s, or M is chosen.

Positive predictive closure. The normalized image of every input in D lies in K off the common null sets. Likelihoods are strictly positive on D. The operator norms of the raw finite-dimensional maps are integrable. Every row $\mu_{a,u}$ is weakly continuous in $u \in D$, and

$$\mathrm{TV}(\ell_a(\cdot,u)\lambda_a, \ell_a(\cdot,v)\lambda_a) \le L_{\mathrm{rep}}\|u-v\|. \tag{6}$$

No image condition on an unrepresented infinite-dimensional state is being inferred.

Actual mass and global cells. For every $a \in A, u \in D$,

$$\mu_{a,u} \le \Lambda \operatorname{vol}_d|_K. \tag{7}$$

Use common finite Borel grid partitions $\mathcal{P}_m$ of K, with at most m cells, compact convex cell closures and mesh at most $C_K m^{-1/d}$. Their boundaries have zero mass under every row. A bounding-cube grid intersected with convex K supplies these partitions; the claimed density is an actual pushforward density, not a distribution assigned to K by hand.

Feedback-uniform raw overlap. There are $b \ge 1$, $\epsilon > 0$ and a probability ζ on K such that

$$\mathrm{Law}_u^{\pi}(X_b) \ge \epsilon\zeta \quad \text{for every } u \in D \text{ and every legal } b\text{-call feedback strategy } \pi. \tag{8}$$

This is a certificate on the true acquired process, not on a quantized observer. The overlap can be verified by a paid positive-mass report word, as in Theorem 6.2. No per-step contraction, observer minorization or stationary initial distribution is assumed.

For any probability ν define $q_M(\nu) = \inf_{r_1,\dots,r_M \in B} \int \min_j \|x - r_j\|^2 \nu(dx)$. For a concave function f and a common partition define its conditional gap

$$\mathcal{J}_{\mathcal{P}}(f,\mu) = \sum_{C\in\mathcal{P}} \mu(C) f\left(\frac{\int_C x\, d\mu}{\mu(C)}\right) - \int f\, d\mu, \tag{9}$$

omitting zero-mass cells. Introduce the discounted potentials

$$V_s = \min_{a\in A} K_a H_s, \qquad H_s = sg + \beta V_s, \qquad \psi_s = H_s/s. \tag{10}$$

Theorem 3.2 (Stopped causal acquired-geometry transfer)**.** *Assume Definition 3.1 and the prepared executable task. The value in* (5) *is the unique continuous solution of* (10)*. There are constants $0 < c \le C < \infty$ depending only on the fixed certificates, not on $u_0 \in D$, $s \in (0,1]$ or M, such that for $M \ge 2$*

$$\mathcal{R}_{\mathrm{cp}}(s,M;u_0) = \inf_{\pi}\{B_s^{\pi} + q_M(\nu_s^{\pi})\}, \tag{11}$$

$$cM^{-2/d} \le \mathcal{R}_{\mathrm{cp}}(s,M;u_0) - V_s(u_0) \le CM^{-2/d}, \tag{12}$$

$$cM^{-2/d} \le \mathcal{R}_{\mathrm{hom}}(s,M;u_0) - V_s(u_0) \le C(M-1)^{-2/d}. \tag{13}$$

The lower bounds hold against all legal randomized Borel finite-label programs, including programs with time-dependent action and update rows. The upper is achieved by one homogeneous program for the given (u_0, s, M), with one initial label and at most $m = M - 1$ recurrent labels.

More precisely, that program has a normalized terminal ensemble $Z = (z_j)$, with positive-mass means q_j, and a frozen recurrent instrument P satisfying

$$Z = sB_0 + \beta PZ = s\sum_{t\ge 0} \beta^t P^t B_0. \tag{14}$$

Here B_0 is the true ensemble produced by its first acquisition from u_0, not an invariant preparation. Its actual pooled acquired law is

$$\mu_s = s\mu_{a_0,u_0} + \beta \sum_{i,a} e(z_i) p_{ia} \mu_{a,q_i}. \tag{15}$$

Its full excess risk has the exact representation

$$R(\gamma_{s,m}; u_0) - V_s(u_0) = \mathcal{J}_{\mathcal{P}_m}(\psi_s, \mu_s). \tag{16}$$

The measure in this equality is generated by the executed table and its preparation-injected occupation, not a chosen stationary law. The unique resolvent solution does not assert uniqueness of the nonlinear calibration fixed point or convergence of an algorithm for finding it.

The exact construction and (16) use closure, common null boundaries and finite actions; overlap is used only for the uniform Lipschitz bound on ψ_s. Without (8), the same identity and an upper with the actual finite constant $\operatorname{Lip}_K \psi_s$ remain valid for each fixed $s > 0$. Thus the theorem separates causal realizability from a uniform resource exponent. The uniform density quantifier covers both full-history predictive states and means created by pooling: the latter need not lie on a particular prepared trajectory.

Corollary 3.3 (Internally implemented geometric stopping)**.** *For $M \geq 3$, under the same hypotheses,*

$$cM^{-2/d} \leq \mathcal{R}_{\mathrm{int}}(s, M; u_0) - V_s(u_0) \leq C \left\lfloor \frac{M-1}{2} \right\rfloor^{-2/d}.$$

The construction uses one initial nonhalt label, m recurrent nonhalt labels and m halt labels. Its independent fresh stop coin and halt status are part of the implementation; it does not store an elapsed-time counter. Its mean and pathwise call budgets are those specified after (4)*.*

4. Proof of the central theorem

4.1. Controlled Bellman oscillation, not observer mixing.

Lemma 4.1 (Uniform stopped Bellman potential)**.** *The function V_s in* (10) *is concave. Writing $U_s = V_s/s$,*

$$\operatorname{osc}_D U_s \leq \frac{b \operatorname{osc}_D g}{\epsilon}, \qquad \operatorname{Lip}_K \psi_s \leq L_* := \operatorname{Lip}_K g + \frac{b \operatorname{osc}_D g}{\epsilon r}. \tag{17}$$

Moreover $\psi_s - g$ is concave. Without overlap one may replace the oscillation bound by $\operatorname{osc}_D g/s$.

Proof. The map $Tf = \min_a K_a(sg + \beta f)$ is a sup-norm contraction with factor $\beta < 1$ (constant when $s = 1$). Its iterates from zero converge uniformly to a continuous fixed point. The usual finite backward induction and the bounded remainder $\beta^t L_0$ identify it with the infimum in (5). A Borel minimizing action selector exists because the action set is finite. Thus this infimum ranges over all full-history adaptive policies, not merely policies selected in advance without feedback.

If f is concave on D, its perspective $wf(v/w)$ is concave on the cone over D. Since the normalized output of each raw instrument lies in K, integration in (1) makes $K_a f$ concave on D. A finite minimum of concave functions is concave. Uniform iteration proves concavity of V_s and hence of $\psi_s - g = \beta U_s$.

For $\mathcal{T}_\beta f = \min_a K_a(g + \beta f)$, compare the same b-call feedback policy when its terminal function is f or q. Running terms cancel. The difference of the optimal values is bounded above and below by β^b times expectations of $f - q$ under legal terminal laws. By (8), every such law is $\epsilon\zeta + (1-\epsilon)\widetilde{\mu}$. Therefore

$$\operatorname{osc}_D(\mathcal{T}_\beta^b f - \mathcal{T}_\beta^b q) \leq \beta^b (1-\epsilon) \operatorname{osc}_D(f - q).$$

Since $U_s = \mathcal{T}_\beta^b U_s$ and $\operatorname{osc}_D(\mathcal{T}_\beta^b 0) \le b \operatorname{osc}_D g$, rearrangement gives $\operatorname{osc}_D U_s \le b \operatorname{osc}_D g/[1-\beta^b(1-\epsilon)]$, which implies (17). The crude bound follows by summing the discounted ranges of g. For a concave function on D, extending a line through K by length r in either direction bounds intermediate slopes by its oscillation divided by r. Apply this to U_s. No observer transition matrix appears in this proof. □

4.2. **Acquired mass and nonsmooth curvature.**

Lemma 4.2 (Common-cell curvature and all-center converse)**.** *If $\mu \le \Lambda \operatorname{vol}_d|_K$ and f is concave and L-Lipschitz on K, the bounding-cube grids satisfy*

$$0 \le \mathcal{J}_{\mathcal{P}_m}(f,\mu) \le C_{K,d}\Lambda L m^{-2/d}. \tag{18}$$

Every such μ also satisfies

$$q_M(\mu) \ge \tfrac{1}{2}(2\Lambda v_d M)^{-2/d}, \tag{19}$$

where v_d is the unit-ball volume. Both estimates survive mixing with the original probability weights.

Proof. We recall the acquired-curvature argument to include nonsmooth action switches and boundary cells. The convex function $-f$ has a convex L-Lipschitz extension to $\mathbb{R}^d$: take the supremum of supporting affine functions at interior points of K and use continuity on its boundary. Its distributional Laplacian κ is a positive measure. For a cube of side h and center z, mollify the extension and subtract its tangent plane at z, obtaining a nonnegative convex function w with $w(z) = 0$. On the surrounding ball, the submean inequality bounds w on the cube by a dimensional constant times its spherical average. Convex radial slopes satisfy $w(z + R\theta) \le R\partial_R w(z + R\theta)$. Integrate this inequality and apply the divergence theorem. After removing mollification on a slightly enlarged ball it gives

$$\int_C (-f)\, d\mu - \mu(C)(-f)(\mathbb{E}[X \mid C]) \le C_d h^{2-d}\mu(C)\kappa(\overline{B}_{3\sqrt{d}h}(z)).$$

Affine terms cancel at the centroid. In dimension one the sphere is its pair of endpoints and the same inequality follows by the two one-sided derivatives. A smooth cutoff on a fixed neighborhood of K bounds the total curvature there by $C_K L$, through integration by parts. The enlarged balls have bounded overlap and $\mu(C) \le \Lambda h^d$. Summation proves $C\Lambda L h^2$. Dividing a fixed bounding cube into $\lfloor m^{1/d} \rfloor$ parts per coordinate gives at most m cells and proves (18), including coarse meshes by enlarging the fixed constant. No smooth Hessian or separation of acquired mass from ∂K was assumed.

For the converse, M balls of radius t about arbitrary readouts have mass at most $\Lambda M v_d t^d$. At $t = (2\Lambda M v_d)^{-1/d}$ at least half the actual mass lies outside. Randomized assignments cannot improve the pointwise nearest-readout loss. This proves (19) also for off-support centers and repeated readouts. □

4.3. **Preparation-injected calibration.**

Lemma 4.3 (A fixed table calibrated at the stopping call)**.** *For every common grid there is a homogeneous table with one initial label and its grid labels, whose terminal ensemble is* (14)*. At each positive-mass recurrent label its programmed q_i is the exact conditional mean at the geometric stopping call, and its action row is supported on* $\arg\min_a K_a H_s(q_i)$.

Proof. Let D_j be the compact convex closure of cell C_j. Work on the compact convex normalized ensemble base

$$\mathcal{E}_m = \{(z_j) : z_j \in C_{D_j},\ \sum_j e(z_j) = 1\}.$$

The initial label is never revisited. Choose any minimizing action a_0 in (10) at u_0 and define the genuinely acquired first-call ensemble

$$(B_0)_j = \int \mathbf{1}_{\{F_{a_0}(y,u_0)\in C_j\}} L_{a_0}(y)(1,u_0)\, d\lambda_{a_0}(y).$$

For $z_i = w_i(1,q_i)$ with $w_i > 0$, permit any action probability supported on the finite minimizing set at q_i. Send report y to the cell containing $F_a(y,q_i)$. At zero-mass rows choose any representative in D_i; its contribution is zero. The correspondence is

$$\mathcal{F}_j(Z) = s(B_0)_j + \beta \sum_{i,a} p_{ia} \int \mathbf{1}_{\{F_a(y,q_i)\in C_j\}} L_a(y) z_i\, d\lambda_a(y). \tag{20}$$

Every value lies in $\mathcal{E}_m$: each positive contribution is the actual likelihood-weighted barycenter of predictions in its output cell. The values are nonempty compact convex, since for a fixed Z the action simplices enter linearly.

Its graph is closed. For a positive limiting row mass, representatives converge; a subsequence of its action probabilities converges to a distribution on limiting minimizers, by continuity of $K_a H_s$. The normalized map is continuous in its incoming coordinates off the common null sets. Its boundary preimage has zero reference mass, because the likelihood is positive and its acquired row charges no grid boundary. Dominated convergence with the integrable operator norm proves convergence of the indicator integrals. At a row whose mass tends to zero, every contribution is bounded by that mass in the cone base norm; no normalization at zero is used. Kakutani's theorem [1] now gives a fixed point Z and witnessing action rows. Fix these rows and choose any within-cell representatives and action rows at zero-mass labels. This defines an ordinary Borel table at every label.

For that frozen table define the linear recurrent map P by the integral part of (20), using these fixed representatives and action probabilities on every input ensemble. Positive input ensembles over K remain over K; their total mass is preserved. Hence P contracts the sum base norm, and $I - \beta P$ has inverse $\sum_{t\geq 0} \beta^t P^t$. The fixed-point equation identifies its unique terminal solution as (14).

Starting the actual machine at the initial label gives B_0 after one report and $P^{t-1}B_0$ after t reports on its unaudited continuation. Independent stopping therefore gives exactly $s\sum_{t\geq 1} \beta^{t-1} P^{t-1} B_0 = Z$. Thus the q_i are calibrated at the actual stopping call. They are not asserted to equal $\mathbb{E}[X_t \mid S_t = i]$ for each deterministic t. The preparation term identifies the first run even when P has several invariant ensembles or is periodic. For $s = 1$ the correspondence is constant and the same argument includes the one-call task. □

Lemma 4.4 (Exact stopped regret)**.** *The table of Lemma 4.3 satisfies* (16)*.*

Proof. The probability μ_s in (15) is the likelihood-weighted mixture of the initial row with weight s and the calibrated recurrent rows with weights $\beta w_i p_{ia}$. The fixed-point equations say that $\mu_s(C_j) = w_j$ and its cell centroid is q_j. The true stopped response risk is $\sum_j w_j g(q_j)$ by the affine audit formula and terminal calibration.

Bellman-minimizing action rows give

$$\int H_s\, d\mu_s = sV_s(u_0) + \beta \sum_i w_i V_s(q_i),$$

whereas

$$\sum_j w_j H_s(q_j) = sR(\gamma_{s,m}; u_0) + \beta \sum_j w_j V_s(q_j).$$

Subtract and divide by s. This is precisely (16). In particular, no stationarity replacement or optional-stopping substitution in a fixed-time identity has been made. A single incoming row may

have an optimal static quantizer without satisfying (20); the common cells and global fixed point are essential. □

4.4. **Matching risk and all-program quantifiers.** For the constructed table, (7) passes to μ_s. Lemmas 4.1 and 4.2 applied to (16) give $C(M-1)^{-2/d}$. Strong concavity of g and concavity of $\psi_s - g$ also imply that its Jensen gap is at least the actual within-cell squared error, although this alone is not the all-program converse.

For an arbitrary legal program, condition the protected audit on its full pre-audit history, including its private coins. Its risk is

$$B_s^\pi + \mathbb{E}\|X_\tau - r_{S_\tau}\|^2.$$

The decoder has at most M distinct responses. The independent hazard makes ν_s^π a mixture of actual rows, so (19) bounds the second term below by $cM^{-2/d}$. The first term is at least $V_s(u_0)$ even when the action strategy is time-dependent. This proves the lower bounds. For a checkpoint, conditional nearest-center assignment gives (11); an infimum not attained can be approached by measurable finite-center codes. A global grid after an optimal full-history acquisition selector gives its upper. These arguments prove Theorem 3.2 without assuming that the finite program's acquisition strategy coincides with the full-history optimal one.

5. Stopping resources, hard caps and acquired strata

5.1. **The independent stopping device is not free memory.**

Proof of Corollary 3.3. Take $m = \lfloor (M-1)/2 \rfloor$ cells in Theorem 3.2. The initial label executes its programmed first action. At every acquired report compute the same destination cell as that table, and use an independent fresh Bernoulli coin. On continuation enter that cell's recurrent copy; on termination enter its halt copy and invoke the protected audit. The total is $1 + 2m \le M$. Each halt copy stores the same readout as its recurrent copy. The stopped physical–label law is unchanged, so the upper follows. Conversely the imposed history-independent hazard preserves the actual row mixture at the stop, and any internally stopped M-state program still has at most M terminal readouts. The all-program projection proof gives its lower. The signal to invoke the audit comes from the halt label, not from an external countdown. □

Architecture	Initial	Recurrent	Halt / clock
External geometric request	1	m	Auditor's fresh coin
Internal geometric stop	1	m	m halt labels
External capped request	1	m	Auditor's coin and cap counter

All rows use τ genuine raw acquisitions and one preparation and audit. Generating a noncomputable real hazard is not a finite-bit primitive: Section 9 treats the exact hazard as a specified randomization primitive, or a computable hazard with separately charged implementation error. No claim identifies a label count with total bit complexity.

Theorem 5.1 (A separately proved hard-cap transfer)**.** *Fix $H \ge 1$ and replace the external geometric request by $\tau_H = \min(\tau, H)$. The auditor keeps its own cap counter, not visible to the controller before requesting the response. Let $V_{s,H}(u_0)$ be the full-history acquisition optimum for this task. With $M \ge 2$,*

$$cM^{-2/d} \le \mathcal{R}^{\mathrm{cap}}_{\mathrm{hom}}(s, H, M; u_0) - V_{s,H}(u_0) \le C(M-1)^{-2/d} + 2L_0\beta^H. \tag{21}$$

The upper uses the same frozen table constructed for the uncapped hazard. The cap uses at most $N = H + 2$ physical calls and mean raw acquisitions $(1-\beta^H)/s$. If $\beta^H \le M^{-2/d}$ the upper and lower match uniformly. For $s < 1$ it suffices that $H \ge (2\log M)/(d[-\log(1-s)])$; for $s = 1$ the tail is zero.

Proof. For any fixed pre-request policy and readout, couple the two tasks with the same raw experiment, actions and coins. They agree on $\{\tau \le H\}$. On its complement each bounded audit loss lies in $[0, L_0]$, so risks differ by at most $L_0\beta^H$. Policies may be extended beyond H arbitrarily, which gives the same bound between the full-history infima. Apply these two bounds to Theorem 3.2. At every possible capped cut the actual predictive law is still a mixture of the stipulated rows; the small-ball argument relative to the policy's own Bayes risk proves the lower without subtracting a tail term. Finally sum the survival probabilities $\sum_{t=1}^{H} \mathbb{P}(\tau \ge t)$. An internally counted cap is a different architecture and would require counting its phase memory; it is not granted by this theorem. □

Theorem 5.2 (Marked acquired-rank law)**.** *Replace* (7) *by a finite visible decomposition. Each report carries a mark j whose actual conditional probability is the fixed $w_j > 0$, independent of the incoming state and chosen action. Given this mark, the acquired vector lies in a compact convex affine set $K_j \subset K$ of dimension $d_j \ge 1$, with relative density at most Λ_j on that span, uniformly on D. Separately marked atom states are allowed; at least one component has positive dimension. Keep closure, continuity and the feedback-overlap certificate on the predictive coordinate. Relative grid boundaries are null. Put*

$$\Xi(M) = \sum_{j:d_j>0} w_j M^{-2/d_j}. \tag{22}$$

For all M above a fixed component-dependent threshold, the checkpoint, homogeneous and internally stopped excess risks are comparable to $\Xi(M)$, uniformly over $s \in (0, 1]$ and $u_0 \in D$. The capped upper is $C\Xi(M) + 2L_0\beta^H$. Constants may depend on the fixed positive weights and components; no optimal allocation as the weights vary with M is claimed.

Proof. Reserve one recurrent label for each atom and divide the remaining labels in fixed positive proportions among the positive-dimensional marked grids. Use their compact relative cell cones in (20). An output carries its actual mark, so each raw contribution belongs to its designated relative cell cone. Both the first-call injection and the recurrent rows retain the original w_j. Closed-graph and zero-mass arguments are unchanged, using relative rather than ambient boundary nullity. The same positive resolvent identifies the actual terminal ensemble, even when a mean entering an instrument is off the stratum of an earlier report.

The Bellman function is concave and Lipschitz on ambient K, so its restriction to each affine span has the same Lipschitz bound. Apply Lemma 4.2 on each positive-dimensional component and sum its conditional gap with multiplier w_j. Atoms have zero cell gap. This proves the upper $C\sum_j w_j m_j^{-2/d_j} \le C\Xi(M)$. For any global M-center decoder, orthogonally project its centers onto each affine span. This cannot increase distances. The relative small-ball bound gives $c_j M^{-2/d_j}$ there; multiply by the actual w_j and sum. The stopped row mixture has these same weights for every acquisition policy because the prescribed hazard is independent. This proves all lower bounds, including for overlapping affine spans and off-stratum readouts. Internal halt copies change counts only by a fixed factor. The cap proof applies unchanged. No component is conditioned on and then assigned unit mass. □

6. Two raw controlled realizations

The following proofs verify the mother theorem independently. They do not infer the general statement from a model-specific rate. Their density and accessibility mechanisms were developed in the retained controlled and finite-acquisition articles [12, 13]; we give the verifications here because the stopped law uses every hypothesis, including the preparation domain.

Theorem 6.1 (Controlled Gaussian hidden-state experiments)**.** *Let the hidden physical state have $q \ge 2$ values. Let P be an invertible column-stochastic matrix with $P_{ij} \ge a_0 > 0$. Before the*

transition, a chosen sensor a reports a Gaussian vector with mean m_i^a conditional on hidden state i, covariance $\Sigma_a > 0$, and report dimension $q-1$. The finite sensor set has full-rank likelihood-difference matrices. The hidden state then transitions by P. A protected categorical audit of the current state is scored by squared probability loss. All conclusions of Theorems 3.2 and 5.1, and Corollary 3.3, hold with $d = q-1$. The comparison includes every full-history adaptive sensor policy, not just a fixed sensor word.

Proof. The categorical audit separates the probability vector. Every legal future instrument-word probability is linear in this vector, yielding the finite predictive quotient and its positive simplex cone directly from the raw hidden-state kernel. All simplex preparations and their mixtures are allowed. Relative to Lebesgue measure the raw instrument is

$$L_a(y)u = P\,\mathrm{diag}(f_1^a(y),\dots,f_q^a(y))u.$$

A positive normalized mixture of the finitely many Gaussian densities can instead be used as reference probability, with the corresponding likelihood ratio. This changes no acquired law. The order is report first, transition second.

Take $K = P\Delta_{q-1}$ and $D = \{u \in \Delta_{q-1} : u_i \geq a_0/2\}$. Relative to the simplex hyperplane, K is full-dimensional, compact and contained in the interior of D. All raw outputs are in K, and each likelihood is positive on D. Linearity gives report total-variation continuity; dominated integration gives weak row continuity. The loss is $g(u) = 1 - \|u\|^2$ on the probability hyperplane, with its actual categorical variance.

Before applying P, let w be the normalized Bayes vector and $z_i = \log(w_i/w_q)$. Then

$$z_i = \log(u_i/u_q) + (m_i^a - m_q^a)^T\Sigma_a^{-1}y - \tfrac{1}{2}\left[(m_i^a)^T\Sigma_a^{-1}m_i^a - (m_q^a)^T\Sigma_a^{-1}m_q^a\right].$$

Under the true report law this is a finite mixture of nonsingular Gaussian vectors. Compact D, finitely many actions, positive covariance eigenvalue bounds and the nonzero likelihood-rank determinants give a uniform density bound $C\exp(-c\|z\|^2)$. The inverse logistic Jacobian is a fixed multiple of $(\prod_i w_i)^{-1}$, bounded by $C\exp(C\|z\|)$. Thus Gaussian decay dominates the Jacobian on the entire simplex, including near every boundary face. The invertibility of P preserves a uniform upper density on K.

On a fixed smaller interior simplex W_*, the same Gaussian-mixture formula gives a uniform strictly positive density lower for every $u \in D$ and sensor. Its image PW_* supports a common normalized-volume measure ζ with $\mu_{a,u} \geq \epsilon\zeta$. Hence (8) holds with $b = 1$ even for adaptive policies. Relative hyperplane grids have null boundaries. This proves every certificate; no tail, unsuccessful report or acquisition cost was discarded. Table means are programmed constants for the known preparation and hazard, not a free exact filter register. □

Theorem 6.2 (Noncommuting controlled quantum experiments)**.** *Let $D_q \geq 2$, $0 < \eta, \lambda < 1$ and a finite nonempty family of unitaries U_a be fixed. On the unit ball of traceless Hermitian matrices, with normalized Hilbert–Schmidt volume λ_0, put $E_H = I + \eta H$, $K_H = \sqrt{E_H}$, $\tau_* = I/D_q$, and*

$$L_a(H)\sigma = \lambda U_a K_H \sigma K_H U_a^* + (1-\lambda)\,\mathrm{tr}(E_H\sigma)\tau_*. \tag{23}$$

The action is chosen before H. Use a fixed informationally complete finite positive-operator-valued measurement for the protected categorical audit. Theorems 3.2 and 5.1, and Corollary 3.3, hold in its induced affine probability chart with $d = D_q^2 - 1$. The observer has classical labels only; no quantum memory is provided. Constants may deteriorate near degenerate endpoints of η, λ.

Proof. The effects integrate to I and satisfy $(1-\eta)I \leq E_H \leq (1+\eta)I$. Thus (23) is a completely positive normalized instrument: an unsharp measurement, a controlled unitary, and depolarization. Its report density $1 + \eta\,\mathrm{tr}(H\sigma)$ depends on the incoming state. This is a repeated-measurement instrument in the operational sense [2]; the unsharp effects generally do not commute.

Informational completeness separates density operators. Every future instrument-word expectation is affine in the density operator. The physical density base and positive cone therefore give the stipulated predictive realization and a Borel preparation section. The audit chart is injective affine; its Euclidean Brier metric is equivalent, with fixed constants, to Hilbert–Schmidt distance. The actual sampling variance is not removed.

Let $K = \{\sigma \geq (1-\lambda)\tau_* : \operatorname{tr}\sigma = 1\}$ and $D = \{\sigma \geq (1-\lambda)\tau_*/2 : \operatorname{tr}\sigma = 1\}$. All normalized raw outputs lie in K, even from a rank-deficient physical input. There is a compact relative buffer. Positivity, linearity and the compact effect domain give all domination and report-continuity conditions.

For fixed interior ρ, the map $E \mapsto \sqrt{E}\rho\sqrt{E}/\operatorname{tr}(E\rho)$ on trace-D_q positive effects has inverse

$$B_{\rho,\sigma} = \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}, \qquad E = \frac{D_q B_{\rho,\sigma}^2}{\operatorname{tr} B_{\rho,\sigma}^2}.$$

Indeed the positive solution of $B\rho B = \sigma$ is unique; substituting it proves both inverse identities. On $\rho \in D$ and the effect ball, eigenvalues have positive lower bounds, so the inverse derivative is uniformly bounded. The forward Jacobian is consequently bounded away from zero. The final depolarizing/unitary map is invertible on the trace-one hyperplane. The bounded actual report density then gives (7) in dimension $D_q^2 - 1$.

At $\rho = \tau_*$ the output is $\tau_* + (\lambda\eta/D_q)U_a H U_a^*$. An interior effect ball covers a common neighborhood of τ_*. Smooth inversion and finitely many actions imply a local actual overlap $\mu_{a,q} \geq \epsilon_0\zeta$ for every q in a sufficiently small closed neighborhood G of τ_*.

This local overlap is reached under every adaptive policy. With $W(\rho) = \|\rho - \tau_*\|_{\mathrm{HS}}$, the matrix square-root bound and likelihood floor give

$$W(F_a(H,\rho)) \leq \lambda W(\rho) + C\lambda\eta t \quad (\|H\|_{\mathrm{HS}} \leq t,\ \rho \in K).$$

One obtains the estimate by expanding numerator and denominator, using $\|\sqrt{I+\eta H} - I\|_{\mathrm{op}} \leq C\eta t$ and $\operatorname{tr}(E_H\rho) \geq 1 - \eta t$. A quiet report ball has actual conditional probability at least $(1-\eta)t^d$, for every incoming state and action. Choose t so that the drift floor is smaller than one quarter of the radius of G, and choose k so that $\lambda^k \sup_K W$ is smaller than another quarter. After the first report has entered K, k consecutive quiet reports bring the exact predictive state into G, regardless of all feedback choices. The next row contributes the local overlap. Its genuine subprobability weight is at least $\epsilon_0[(1-\eta)t^d]^k$. Thus (8) holds with $b = k+2$. Every report in this argument is paid; restricting to a report word is only a mass witness, not a postselected new experiment. The general stopped theorem now applies. No infinite-dimensional operator closure is used. □

Corollary 6.3 (Raw changing acquired rank)**.** *After the instrument in Theorem 6.2, independently select a visible mark j with probability w_j and apply a trace-preserving unital conditional expectation $\mathcal{E}_j$ onto a matrix subalgebra. Include the identity with positive weight; allow complete depolarization as an atom. Then Theorem 5.2 applies, with d_j the affine dimension of the trace-one part of that subalgebra.*

Proof. The raw marked instrument is $w_j\mathcal{E}_j L_a(H)$. It is positive and normalized after summing marks. Each expectation fixes τ_* and preserves the positive spectral floor. In Hilbert–Schmidt coordinates it is an orthogonal projection. Fubini's theorem projects a bounded density on compact K to a bounded relative density; the bound is the original bound times a maximal compact-fiber volume. A fixed audit chart changes only constants. Complete depolarization gives the exact atom τ_*.

For feedback overlap, restrict the quiet report word and its final overlapping report to the identity mark. Multiply the genuine mass witness by w_{id}^{k+1}. All other marks continue to execute on the complementary event with their unchanged weights. Relative grids provide the required cells. This is a change of acquired predictive rank, for example between full density coordinates and diagonal

coordinates. The physical matrices stay strictly positive; it is not a claim of eigenvalue-rank change or a theorem about arbitrary pure-state boundaries. □

7. An explicit first-run binary code

Proposition 7.1 (Preparation-dependent Gaussian calibration)**.** *Take hidden states* ± 1*, reports* $Y \mid i \sim N(i,1)$*, transition matrix with diagonal* $3/4$ *and off-diagonal* $1/4$*, and initial probability* $u_0 = 1/2$*. The scalar predictive coordinate is the probability of the positive hidden state. Let* Φ *denote the standard Gaussian distribution function and put*

$$G(q) = \frac{1}{4} + \frac{1}{2}\{q\Phi(1+\ell) + (1-q)\Phi(1-\ell)\}, \qquad \ell = \tfrac{1}{2}\log\frac{q}{1-q}. \tag{24}$$

For each $s \in (0,1]$*, the equation*

$$q_s = sG(1/2) + (1-s)G(q_s) \tag{25}$$

has a unique solution on $[1/2, 3/4]$*. A homogeneous externally stopped three-label program attains the exact categorical Brier risk*

$$R_s = 2q_s(1-q_s). \tag{26}$$

Five labels implement the same law with internal halting. The code and its risk are explicit; no claim is made that three or five labels are globally optimal for this benchmark.

Proof. Use the initial label with programmed belief $1/2$, and two recurrent labels with beliefs q and $1-q$. At an incoming belief p send the report to the upper label precisely when $y \geq -\frac{1}{2}\log(p/(1-p))$. This is the common cut at posterior probability $1/2$ after the transition. Symmetry gives equal mass $1/2$ to the two labels in the terminal ensemble. For a symmetric incoming ensemble with means $q, 1-q$, direct Gaussian integration gives conditional upper output mean $G(q)$; the first report from preparation $1/2$ gives $G(1/2)$. The resolvent calibration is therefore exactly (25), not the invariant equation $q = G(q)$.

Write ϕ for the Gaussian density. The identity $q\phi(1+\ell) = (1-q)\phi(1-\ell)$ cancels the derivative of the threshold and yields

$$G'(q) = \tfrac{1}{2}[\Phi(1+\ell) - \Phi(1-\ell)], \qquad 0 \leq G'(q) \leq \tfrac{1}{2}.$$

The map in (25) preserves $[1/2, 3/4]$ and contracts it by at most $(1-s)/2$. Its unique fixed point identifies the actual stopped conditional means. The categorical loss is $2p(1-p)$, which proves (26). Iteration $q_{k+1} = sG(1/2) + (1-s)G(q_k)$ has the rigorous a posteriori error bound

$$|q_k - q_s| \leq \frac{[(1-s)/2]^k}{1-(1-s)/2}|q_1 - q_0|.$$

The internal construction duplicates the two output labels into continuing and halting copies and counts its initial label.

For explicit geometry constants use $D = [1/8, 7/8]$, $K = [1/4, 3/4]$, and $A = 2 + \log 7$. The log-odds density is a mixture of variance-four Gaussians with means of magnitude at most A. Since the derivative of inverse logistic followed by the transition is at most $8e^{|z|}$, a scalar acquired-density upper is

$$\Lambda = \frac{4}{\sqrt{2\pi}}e^{A+2}.$$

On $|z| \leq \log 3$ the mixture density is at least $(2\sqrt{2\pi})^{-1}\exp[-(A+\log 3)^2/8]$, and the inverse Jacobian is at least 8. Consequently every row dominates ϵ times uniform probability on $[3/8, 5/8]$, where

$$\epsilon = \frac{1}{\sqrt{2\pi}}\exp[-(A+\log 3)^2/8] > 0.$$

These are conservative explicit raw constants, valid without truncation. Scalar versus probability-hyperplane norms change fixed factors only. Thus the benchmark verifies the center theorem as well as giving a concrete table. A full-history checkpoint with two terminal labels can simulate this table's two-valued terminal readout, so its optimal risk is no larger than R_s; no unsupported formula for its optimal two-center partition is asserted. □

8. Common-task error and resource composition

A simulation compares experiments with the same protected audit and committed response, not their separate full-history baselines. This distinction is needed even under squared loss.

Proposition 8.1 (Typed stopped simulation)**.** *Let a parameter-independent causal simulator lift every legal program of a target experiment to a source experiment. It must specify action compatibility, causal report transformation, its internal state, the strategy lift, the stopping interface, the protected audit and the task readout. Suppose, for every lifted program, the complete stopped law differs from the target law by total variation at most ε and the common loss lies in $[0, L]$. If the target program has M labels and the simulator has at most R retained states, its direct source realization uses at most RM labels and satisfies*

$$R_{\mathrm{source}}(\widetilde{\gamma}) \le R_{\mathrm{target}}(\gamma) + L\varepsilon.$$

If one target call uses at most c source acquisitions, including unsuccessful attempts, and initialization uses c_0, then the source pathwise count is at most $c_0 + c\tau$ and its mean is at most $c_0 + c/s$. Program, workspace, calibration and physical time remain separate accounts. Composing two such simulators adds their uniform complete-law deficiencies and multiplies their state counts.

Proof. The product of target and simulator state supplies the next source action, report interpretation and halt status; no target history outside that state is retained. Couple the complete laws maximally. Their losses agree off a set of probability at most ε, proving the risk estimate. Pathwise counting precedes expectation. For composition, apply the triangle inequality and the contraction of total variation under a parameter-independent Markov kernel, uniformly over the lifted strategies. This comparison includes the audit and stopping randomness: agreement only on each fixed-time report marginal does not establish the asserted complete-law coupling. The argument gives no reverse minimax lower unless an appropriate reverse simulation or a separate hard experiment is proved. □

Theorem 8.2 (Attained same-task calibration and deficiency)**.** *Augment an experiment of Theorem 3.2 by an independent fair bit U. One paid call reports U with probability $1 - a$ and an erasure mark with probability $a \in [0, 1]$. Later core calls contain no additional information about U conditional on the selected core actions. A final protected audit scores the core target, U, and an invisible fixed calibration value $\vartheta \in \{-\delta, \delta\}$ by the sum of squared losses, where $0 \le \delta \le 1$. The program knows a, δ, not U or ϑ.*

The parameter-independent causal deficiency of this interface to the exact early-bit interface, with that bit required before core acquisition and protected at the terminal audit, equals $\varepsilon = a/2$. Let $\mathcal{R}_{\mathrm{joint}}(s, M, \delta, \varepsilon)$ be the infimum worst-case risk over homogeneous observers with an external independent geometric auditor. For $M \ge 7$,

$$\mathcal{R}_{\mathrm{joint}}(s, M, \delta, \varepsilon) - V_s(u_0) \asymp M^{-2/d} + \delta^2 + \varepsilon. \tag{27}$$

The constants are uniform in s, M, a, δ. The pathwise physical acquisition count, including preparation, early bit and terminal audit, is $\tau + 3$, of mean $s^{-1} + 3$. An internal-halt implementation uses $6m + 4$ labels in place of the external construction's $3m + 4$; hence the same order holds for $M \ge 10$. No hard cap is implicit.

Proof. On an erasure, guessing fairly simulates the early exact-bit report. Coupling it to the protected bit makes the complete-law mismatch probability $a/2$. Conversely, condition on the erasure. Any report committed before the core calls is independent of U, so it disagrees with the protected bit with probability $1/2$. This event has zero target probability. The total-variation lower is $a/2$, for every causal simulator permitted by this interface. Additional late core calls cannot recover the already required report.

For the risk lower, place the symmetric prior on the two calibration signs. The observations do not depend on this sign, giving squared Bayes loss at least δ^2. The residual bit variance is $a/4 = \varepsilon/2$ even with full history; conditional on the early bit information, the core's arbitrary adaptive acquisition policy cannot improve on $V_s(u_0)$. Conditional projection of the core audit gives its full-history risk plus finite-readout error. The actual stopped core predictive law is still a mixture of the certified rows, including after policies use the available bit information. The all-center argument therefore adds $cM^{-2/d}$. These lower terms coexist under one preparation and task, rather than coming from incompatible adversarial examples. The worst-case calibration risk dominates their Bayes average.

For the upper, let $m = \lfloor (M-4)/3 \rfloor \geq 1$. Use one initial label for the early-bit action. Its three possible reports select disjoint lanes with bit readouts 0, 1/2, 1. Each lane contains a separate copy of the core initial label and its m recurrent labels. Run the same calibrated core table in all lanes and read out zero for the calibration coordinate. This uses $1 + 3(1+m) = 3m + 4 \leq M$ labels and attains

$$V_s(u_0) + Cm^{-2/d} + a/4 + \delta^2.$$

The lanes are deliberately not merged; their state cost is explicit. Replacing each core copy by its $1 + 2m$ internal-halt construction gives $1 + 3(1 + 2m) = 6m + 4$. The bit call and audit are not free reports. This proves (27) and all claimed accounting. It does not assert that these lane constructions are minimal for each small integer budget. □

An externally capped version follows by imposing τ_H on this same experiment. Proposition 8.1 and Theorem 5.1 add at most a constant times β^H to the displayed excess, and the pathwise count is $H + 3$. Calibration, a numerical tolerance and a deficiency *upper bound* are not interchangeable lower-bound parameters: Theorem 8.2 has explicitly realized the two unavoidable errors.

9. Finite implementation and numerical budgets

Theorem 3.2 is an exact Borel implementation theorem with finitely many retained labels. Its calibrated constants, action probabilities and cell tests are not automatically finite-bit instructions. We next state what certificates suffice to compile them and where the hazard enters numerical sensitivity.

Proposition 9.1 (Stopped perturbation at fixed label count)**.** *Fix a homogeneous table of Theorem 3.2. Suppose a finite implementation can be coupled to it so that, at each call before the first mismatch, the conditional probability of an action, cell, stop or primitive mismatch is at most η. This bound must hold for every true physical predictive input and every programmed row that can occur; it cannot be inferred from calibration only at τ. If its readout error is at most v, and the ideal task diameter is at most D_*, then*

$$|R_{\text{finite}} - R_{\text{ideal}}| \leq L \min\{1, \eta/s\} + 2D_* v + v^2. \tag{28}$$

For the externally capped protocol replace $1/s$ by $(1-\beta^H)/s \leq H$. The implementation uses the same retained labels if all temporary registers are erased before the next call.

Proof. Before the first mismatch the coupled core histories, labels and actions coincide. The chance of reaching call t is β^{t-1} under the ideal independently generated stop process. Sum its conditional mismatch probability over these calls. This gives $\eta \sum_{t \geq 1} \beta^{t-1} = \eta/s$; capping truncates the sum.

On the complement, only the readout changes, and expansion of its squared loss bounds this by $2D_*v + v^2$. The bound uses the actual whole stopped law, not a pointwise asymptotic numerical statement. □

One sufficient per-call certificate decomposes η into primitive-coupling failure, arithmetic failure, action-probability approximation and a cell-boundary term. If the normalized pre-quantization value is perturbed by at most ρ, the last term is bounded by

$$\sup_{x,q,a} \int \mathbf{1}_{\{\mathrm{dist}(F_a(y,q),\partial\mathcal{P}_m)\le\rho\}} \ell_a(y,x)\, d\lambda_a(y). \tag{29}$$

Here x is the true physical predictive input and q the programmed row constant; they need not agree. A common likelihood support, domination and null boundaries imply convergence to zero for each fixed finite table. A computable uniform modulus, when claimed, is an additional numerical certificate. In particular, quantizing a continuous raw report does not make its discrete distribution close in total variation to that raw distribution.

Proposition 9.2 (Effective table approximation)**.** *Fix $s > 0$ and m. Suppose the raw instruments, audit, compact domains and common partitions have effective compact descriptions; their finite instrument integrals and readout costs can be enclosed to arbitrary rational accuracy; and the frozen transition integrals are uniformly continuous in row constants and action probabilities with a supplied computable modulus. Then the optimal risk within the compact family of common-cell tables with one initial label and m recurrent labels is computably approximable from above and below. For any $\xi > 0$, a rationally specified table with the same labels has risk at most*

$$V_s(u_0) + Cm^{-2/d} + \xi.$$

This assertion is finite computability, not polynomial complexity or an effective bound on the search time uniform as $s \downarrow 0$.

Proof. Allow row representatives in all compact cell closures, arbitrary action probabilities, arbitrary initial action probabilities, and readouts in a compact legal response set. This parameter space is compact and contains the table constructed above. For a fixed table, let P and B_0 be its recurrent and first-call instruments. On the sum cone base norm, $\|P\| \le 1$. If another genuine stochastic table has $\|P - \widetilde{P}\| \le e$ and $\|B_0 - \widetilde{B}_0\| \le e_0$, then

$$\|Z - \widetilde{Z}\| \le e_0 + \frac{\beta e}{s}, \quad Z = s(I - \beta P)^{-1} B_0. \tag{30}$$

Subtract the two fixed-point equations and use the Neumann-series norm bound $1/s$ and $\|\widetilde{Z}\| = 1$. Risk is a continuous linear functional of the terminal ensemble for fixed readouts and uniformly continuous in the readouts. Certified integral continuity and (30) therefore give an effective modulus on the compact table space.

Enumerate finite rational nets of this space, enclosing the risk at every net point by rational interval integration and a finite Neumann sum. Its omitted terminal mass is β^T after T calls; its loss remainder is at most $L\beta^T$. Subtract the table-space modulus from the smallest lower endpoint to obtain a global lower bound, and retain the table of smallest upper endpoint for an upper bound. Refinement makes their difference arbitrarily small. No fixed-point solver convergence is assumed, no local optimization failure is used as a lower bound, and the certified minimizing table retains exactly the allowed label architecture. The finite approximants need not preserve exact conditional calibration or exact Bellman tie sets; closeness of the executed risk is what is proved. □

For the Gaussian realization, finite Gaussian mixture integrals, tail enclosures and logistic boundary estimates supply these certificates on compact row domains. For the quantum realization, positive eigenvalue floors and compact semialgebraic effect domains give square-root continuity and integration enclosures; in the marked case the integration is performed separately in each relative

chart. These claims require computable raw coefficients, such as rational parameters and certified computable unitaries. An arbitrary real coefficient is not thereby made finitely describable. The explicit scalar benchmark has the stronger contraction-based compiler of Proposition 7.1.

A finite representation stores $O(m(d+|A|)p)$ bits of row constants, probabilities and responses at precision p, in addition to the read-only primitive code and common partition description. Current report coordinates and arithmetic registers are temporary workspace with an explicit primitive-dependent size $W(p,d)$; they are not persistent across calls. The next label uses $\lceil \log_2(m+1) \rceil$ bits externally, or $\lceil \log_2(2m+1) \rceil$ internally. Program bits are not counted as dynamic labels, but are not declared free: they are a separate resource coordinate. Expected execution cost is at most the certified per-call cost divided by s, plus preparation and audit costs. A hard execution-time cap needs the separate capped protocol. These are sufficient implementation budgets, not matching lower frontiers for program size, arithmetic space or time.

10. Mathematical position and scope

Belief-state reduction, discounted dynamic programming, positive measurement instruments and finite-state control are established tools [2, 3, 5]. Value-directed belief compression [4] selects a predictive representation for task performance. Bayes-risk quantization is related to Bregman divergence [6]. The present conditional curvature is consistent with that interpretation; no priority is claimed for Jensen's inequality, perspective convexity, a Bayesian sufficient statistic or finite-controller approximation in isolation.

Modern filter-stability and finite-memory control results provide near-optimal policies through finite models or windows of past inputs and outputs [7, 8]. Kernel-perturbation bounds [9] analyze model error under discounted and average criteria. The policy-conditional comparison of [10] evaluates finite-window belief mismatch in a fixed closed-loop execution; it is not an all-hard-state converse. Our lower quantifier covers every finite-label observer, while the implemented action table is compared to the full-history adaptive acquisition optimum. The total dynamic alphabet, the finite-window length, an offline belief grid and a free time-dependent controller are different resource objects. These distinctions establish the mathematical statement being compared; they do not by themselves establish historical priority over every theorem in those literatures.

The predictive quotient, normalized Bayesian row decomposition and nonsmooth acquired-curvature estimate are inherited mechanisms [11, 12]. The finite-acquisition article [13] identifies an invariant calibrated table through a raw physical lift and controls entrance from a preparation for a uniform finite audit. The new step here is the preparation-injected fixed point and its positive resolvent: it identifies the *first executed* geometric-stop ensemble, makes its entire regret one Jensen gap, and allows recurrent observer chains with no mixing certificate. The change of audit protocol is explicit. In particular, the removal of an entrance term is not asserted for the same finite-uniform-audit protocol, and one program is not claimed to work without recalibration for every preparation and every hazard.

The two principal raw verifications share the same general theorem. The quantum verification does not assume commuting effects; the Gaussian verification does not truncate reports. The marked extension preserves genuine acquired masses and dimensions, including atoms. These consequences connect the abstract construction to actual probability laws; they do not constitute a general classification of all causal experiments.

The finite affine closure, global legal cells, row-density bound and raw feedback overlap remain substantive sufficient conditions. General conditional expectation is not automatically a realizable finite predictive state. Arbitrary stopping selected from the hidden prediction error, unknown-kernel learning, efficient table synthesis, degenerate or infinitely many acquired strata, and matching joint lower bounds for program/workspace/time are not consequences of the theorem. The imposed independent hazard can be implemented internally, but its optimal choice and general optimal stopping

are separate problems. No total-variation estimate is promoted to large-deviation equivalence, and no finite-dimensional instrument argument is promoted to an unbounded-operator closure.

Appendix A. Boundary cases and the role of the hypotheses

Proposition A.1 (Three noninterchangeable ensembles)**.** *An invariant ensemble of a frozen table, the actual ensemble at a fixed call, and the preparation-injected terminal ensemble need not coincide. Rowwise static quantization is also insufficient for global finite-state calibration.*

Proof. For the identity transition on two labels, both point masses are invariant. Starting at the first label never realizes the second invariant law; its injected geometric resolvent remains the first point mass for every $s > 0$. For a deterministic two-cycle started at the first label, the geometric terminal weights after a first call are $1/(1+\beta)$ and $\beta/(1+\beta)$, whereas the invariant law has equal weights and fixed-call laws alternate. These elementary examples do not satisfy the smoothing theorem's density assumption; they isolate the invalid initialization inference that the general construction avoids.

For the pooling assertion, take two equally weighted incoming rows, each supported on one scalar prediction, 0 or 1. With one output in each row their static distortions are zero. Sending both rows to the same global label gives conditional mean $1/2$ and squared error $1/4$. Local row minimizers do not provide a compatible global program. In the present proof the fixed-point equations hold jointly for all rows and the initial injection. □

A stopping rule that selects reports changes their conditional law. For instance, stop on the first positive report in an independent symmetric sequence; the stopping report is positive with probability one, not distributed as an unconditional row. This explains the conditional independence requirement on the hazard in every all-program lower. The capped independent hazard retains a deterministic-time mixture; arbitrary observation-dependent stopping does not. At $s = 1$ the construction has one raw acquisition and the fixed-point map is constant, so no singular β^{-1} convention is required. At $s \downarrow 0$ the statistical constants remain uniform under raw overlap, while numerical tolerances in (28) must generally become finer. This is the separation between acquired geometry and complete bit/time complexity.

Proposition A.2 (Quantitative separation of two stability certificates)**.** *Let (C) mean a one-step likelihood curvature bound $\frac{1}{2}\Phi'' \le b < 1$ in a scalar predictive coordinate, and (P) mean a strict same-report contraction in a metric bi-Lipschitz to that coordinate on the whole certified domain. Neither certificate implies the other, even for positive instruments with state-dependent report laws.*

Proof. For (P) without (C), use the binary Gaussian instrument of Section 7, now with transition off-diagonal $a = 10^{-4}$ and multiplier $\lambda = 1 - 2a$. Bayes updating adds $2y$ to log odds, so is an isometry in the scalar Hilbert metric. Direct differentiation of the log-odds action of the transition bounds its derivative by $\lambda < 1$. On the compact interval $[a, 1-a]$ this metric is bi-Lipschitz to the scalar coordinate, proving (P).

At incoming probability $p = a$, the curvature coefficient is

$$b(a) = \lambda^2 \int \frac{f_+(y)^2 f_-(y)^2}{[a f_+(y) + (1-a) f_-(y)]^3}\, dy.$$

Restrict y to $[0,1]$. Since $f_+/f_- = e^{2y}$ and $f_+^2/f_- = e^4 \phi(y-3)$,

$$b(a) \ge \frac{\lambda^2 e^4}{(1+ae^2)^3} \int_2^3 \phi(t)\, dt > 1.$$

For an explicit elementary verification, use $\lambda^2 > 998/1000$, $e^4 > 54$, $e^2 < 8$, and the decreasing-density lower sum $\int_2^3 \phi \ge 16^{-1} \sum_{j=1}^{16} \phi(2 + j/16) > 19/1000$. The last strict bound follows by $\sqrt{2\pi} < 251/100$ and bounding each positive exponential series above by its first 31 terms plus its

geometric tail. These rational inequalities give a lower greater than 1.02. Thus (C) fails on the stated domain, without invalidating (P).

For (C) without (P), let $u \in [0,1]$, $r = 1/8$, and give a two-state instrument a report + with likelihood $r(1+u)$ and posterior $F_+(u) = 2u/(1+u)$. It is the positive diagonal instrument with weights $r, 2r$. The other report has the complementary likelihood and resets the physical preparation to probability $1/2$. Both report likelihoods are strictly positive. The integrated squared-response potential is

$$\Phi(u) = \frac{4ru^2}{1+u} + \frac{1-r(1+u)}{4}, \qquad \tfrac{1}{2}\Phi''(u) = \frac{4r}{(1+u)^3} \le \tfrac{1}{2}.$$

Hence (C) holds. But F_+ fixes zero and has derivative two there. More explicitly, $F_+^k(u) = 2^k u/[1+(2^k-1)u]$. If a metric bounded above and below by positive multiples of $|u-v|$ contracted every report by a factor $\kappa < 1$, compare zero with $u = 2^{-k}$ after k successive positive reports. The final Euclidean distance exceeds $1/2$, while the contracted metric distance is bounded by a constant times $(\kappa/2)^k$, a contradiction. The positive-report word has positive probability and cannot be excluded. This latter atomic experiment isolates the stability issue; it is not claimed to satisfy the full-dimensional acquired-density hypothesis of the main theorem. □

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